Ian Hutchinson

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Experience

Northrop Grumman — Baltimore, Maryland

Principal Software Engineer — *June 2022 – Present*

- Led development and delivery of a multi-ship sensor-fusion platform, coordinating engineering efforts across partner organizations and customer teams to deliver a flight-ready product in under six months.
- Architected a configurable, containerized, multi-threaded C++20 application leveraging GitLab CI/CD, Conan, automated testing, and modern software delivery practices.
- Integrated software across multiple organizations and customer environments, enabling automated nightly deliveries, shared development workflows, and streamlined deployment pipelines.
- Drove modernization of a large legacy C++ codebase by reducing technical debt, simplifying dependencies, improving maintainability, and increasing real-time reliability.
- Served as Scrum Master and technical lead, facilitating Agile execution, mentoring engineers, and coordinating cross-functional development efforts.

Software Engineer — *October 2019 – May 2022*

- Delivered pulse-to-pulse radar operation as a new system capability, increasing radar front-end utilization and collection efficiency.
- Integrated machine learning-based automatic target recognition into SAR surveillance products, enabling automated target detection and classification.
- Developed real-time C++ software for airborne radar systems supporting surveillance, reconnaissance, and target detection missions.
- Collaborated with systems, hardware, and test engineering teams to design, integrate, and validate mission-critical radar capabilities.

Associate Software Engineer — *January 2019 – September 2019*

- Developed embedded software for airborne radar systems supporting surveillance and reconnaissance missions.
- Implemented and tested C++ software components for real-time radar processing applications.
- Supported software integration, troubleshooting, and verification activities throughout the development lifecycle.

Siemens — Atlanta, Georgia

Engineering Leadership Development Program Internship — *May 2018 – August 2018*

- Developed and commissioned PLC and HMI control systems for industrial manufacturing facilities, supporting automation and production operations.
- Collaborated with customers, technicians, and engineers to deploy and validate automation solutions in production environments.

MSA – The Safety Company — Pittsburgh, Pennsylvania

Electrical Engineering Co-Op — *January 2017 – May 2017*

- Developed software for automated test fixtures used to validate industrial gas sensors against safety and performance requirements.
- Analyzed test data and collaborated with engineering teams to improve sensor validation and manufacturing processes.

Education

Georgia Institute of Technology — Atlanta, Georgia

Master of Science in Computer Science — *January 2023 – December 2026 (est.)*

Penn State University — University Park, Pennsylvania

Bachelor of Science in Electrical and Electronics Engineering — *August 2014 – December 2018*

Technical Skills

Languages: C++, C, Python, Java, SQL, Bash
Architecture & Systems: Software Architecture, System Design, Distributed Systems, Microservices, Multithreading, Real-Time Systems, REST APIs
Development & DevOps: Linux, Git, GitLab CI/CD, Docker, Conan, Agile/Scrum, Automated Testing, Continuous Integration
Tools & Libraries: Eigen, SQLite, STL